Review of EV Charger Types and Nomenclatures

Introduction

As electric vehicles (EVs) become increasingly prevalent, understanding the intricacies of their charging infrastructure is crucial. From the simple act of plugging in at home to the complex network of public charging stations, a diverse ecosystem of technologies, standards, and nomenclatures governs how EVs are powered. This lecture delves into the various types of EV chargers, their operational modes, power levels, and the specific connectors and communication protocols that enable seamless energy transfer. By exploring these foundational elements, we gain insight into the current state of EV charging and the ongoing efforts to standardize and optimize this rapidly evolving sector, particularly highlighting the unique landscape in India.

Core Concepts

Term	Definition	Significance
`EVSE`	Electric Vehicle Supply Equipment, the physical apparatus that supplies electrical energy for charging EVs.	It is the interface between the grid and the EV, managing power delivery and communication.
`BMS`	Battery Management System, an electronic system that monitors and manages an EV's battery pack.	Crucial for communicating battery status and charging requirements to the EVSE, ensuring safe and efficient charging.
`PLC`	Power Line Communication, a method of communication that uses existing electrical wiring to transmit data.	Enables high-level communication between EV and EVSE, especially for DC charging, by sharing data over power lines.
	Controller Area Network, a robust vehicle bus standard	Used for communication between EV and EVSE,

Term	Definition	Significance
`CAN Communication`	designed to allow microcontrollers and devices to communicate with each other.	particularly in standards like CHAdeMO and some GB/T systems.
`CCS`	Combined Charging System, a type of EV charging connector that supports both AC and DC charging through a single vehicle inlet.	Reduces inventory for EV manufacturers and offers flexibility for users, widely adopted globally.
`CHAdeMO`	A DC fast-charging standard primarily developed in Japan.	One of the earliest DC charging standards, known for its dedicated DC charging capability and CAN communication.
`AC Charging`	Alternating Current charging, where the EV's onboard charger converts AC from the grid to DC for the battery.	Common for home and slower public charging, as the power conversion happens inside the vehicle.
`DC Charging`	Direct Current charging, where the power conversion from AC to DC happens within the EVSE, delivering DC directly to the battery.	Enables faster charging speeds, typically found in public fast-charging stations due to larger EVSE size.
`Mode 1/2/3/4`	Classifications for EV charging based on the connection method and safety features.	Standardizes the approach to AC (Modes 1-3) and DC (Mode 4) charging, indicating safety and control levels.
`Charging Levels`	Classifications based on voltage and power output, indicating charging speed.	Helps categorize chargers by their power delivery capabilities, varying by region (e.g., India's Level 1, 2, 3).

Detailed Analysis

1. Public Charging Station Architecture and Communication

A public charging station is a complex ecosystem designed to serve multiple electric vehicles. It typically comprises several Electric Vehicle Supply Equipment (EVSEs), which can be either AC or DC, and may feature single or dual charging guns. These EVSEs are connected to the AC grid, often through a dedicated transformer and switchgear systems (e.g., circuit breakers, relays) to ensure safe operation. The input supply is commonly three-phase AC, with voltage levels adjusted to meet EVSE requirements, often 400V three-phase AC.

A critical component is the Central Management System (CMS), which communicates with all EVSEs to log data, monitor operations, and facilitate services like metering and user behavior analysis. Communication between EVSEs and the CMS often uses protocols like OCPP (Open Charge Point Protocol). For DC charging, continuous and high-level communication between the EVSE and the EV's Battery Management System (BMS) is essential. This is achieved through dedicated communication channels, often using Power Line Communication (PLC) in systems like CCS2 or CAN (Controller Area Network) communication in systems like CHAdeMO and some GB/T standards. This communication ensures the EVSE responds appropriately to the vehicle's charging requests, adapting voltage and current as needed.

2. Modes of EV Charging

EV charging is classified into different modes based on the connection method and the level of control and protection offered:

- **Mode 1 (AC Charging)**: This is the simplest method, involving direct connection to a standard home socket (e.g., 5A or 15A) using a basic cable. The power conversion from AC to DC occurs within the vehicle's onboard charger. It's primarily for home, slow, or overnight charging with limited power ratings.
- **Mode 2 (AC Charging)**: Similar to Mode 1, it uses a standard wall socket but incorporates a control unit with integrated protection systems, such as a Residual Current Device (RCD), into the charging cable. This adds a layer of safety and control, making it suitable for home charging with a portable charger.
- **Mode 3 (AC Charging)**: This mode utilizes a dedicated AC EVSE with specific connectors. These EVSEs are installed either at home or in public charging stations. They provide a controlled and safer AC charging

experience, with the onboard charger still performing the AC-to-DC conversion.

- **Mode 4 (DC Charging)**: This is for DC fast charging, employing a dedicated DC EVSE. In this mode, the power conversion from AC to DC happens within the EVSE itself, delivering DC power directly to the vehicle's battery. Mode 4 chargers are typically larger, higher-power units found in public charging stations, requiring dedicated communication between the EVSE and the EV.

3. Levels of EV Charging (Indian Context)

Charging levels categorize chargers based on their power output, which can vary by country due to different grid voltage levels. In India, the charging levels are defined as follows:

- **Level 1 (AC)**:
- Voltage: 230V
- Power: Up to 3.5 kW
- **Level 1 (DC)**:
- Voltage: >= 48V to 72V
- Power: Up to 15 kW (typically for systems supporting up to 200A)
- **Level 2 (AC)**:
- Voltage: 380V to 400V
- Power: Up to 22 kW
- **Level 3 (AC)**:
- Voltage: 200V to 1000V
- Power: Greater than 22 kW
- **Level 3 (DC)**:
- Voltage: 200V to 1000V
- Power: Up to 400 kW

4. AC Charger Plugs and Vehicle Inlet Connectors

Different regions and standards have developed various AC charging connectors:

- **AC Type 1 (SAE J1772)**:
- Origin: US, primarily for single-phase or dual-phase AC distribution systems.
- Pins: Two AC power pins (L1, L2/N), two signal pins (Control Pilot, Proximity Pilot), and a protective earth pin.

- Communication: Uses SAE J1772 signaling protocol (PWM-based, low-level communication).
- Usage: Popular in the US and Japan. Also known as Yazaki connector.
- **AC Type 2 (Mennekes)**:
- Origin: Europe, designed to support both single-phase and three-phase AC charging.
- Pins: L1, L2, L3, Neutral, Protective Earth, Control Pilot, Proximity Pilot.
- Communication: Uses SAE J1772 signaling protocol (PWM-based).
- Standard: Specifications defined in IEC 62196.
- Usage: Adopted in Europe, UK, India, and most of the world (except China, Japan, US).
- **GB/T AC**:
- Origin: China, formulated by the Guobiao Standardization Commission.
- Pins: L1, L2, L3, Neutral, Protective Earth, Control Pilot (CP), Charging Confirmation (CC).
- Supports: Both single-phase and three-phase AC.
- Standard: Connector specifications in GB/T 20234.2.
- **AC001 (India)**:
- Origin: Department of Heavy Industry, Government of India.
- Connector: Uses IEC 60309 industrial plug connector (female on EVSE, male from EV).
- Supports: Single-phase supply, with EVSEs often having three independent outputs, each supporting up to 3.3 kW (total 10 kW).
- Communication: No dedicated communication between EV and EVSE. EVSE communicates with CMS via OCPP.
- Usage: Primarily for early models of three-wheelers and cars with low-voltage batteries in India.

5. DC Charger Plugs and Vehicle Inlet Connectors

DC charging requires more sophisticated connectors and communication due to higher power levels and direct battery interaction:

- **CHAdeMO**:
- Origin: Japan, developed by a consortium of Japanese manufacturers.
- Communication: Uses CAN communication.

- Supports: DC charging only (DC+, DC-, CAN-H, CAN-L, Charge Enable/ Disable, Charge Start/Stop, Ground, Proximity Pilot).
- Usage: Adopted in Japan and other countries.
- **CCS1 (Combined Charging System 1)**:
- Concept: Combines AC Type 1 charging with DC charging using a single vehicle inlet connector. The top part accommodates a Type 1 AC plug, while the full connector (including additional DC pins) is used for DC charging.
- Communication: PLC for DC charging (ISO 15118, DIN spec 70121) and PWM signaling (SAE J1772) for AC charging.
- Standard: Complies with IEC 62196.
- Usage: Predominantly in the US and Japan.
- **CCS2 (Combined Charging System 2)**:
- Concept: Combines AC Type 2 charging with DC charging using a single vehicle inlet connector. The top part accommodates a Type 2 AC plug, and the full connector (including additional DC pins) is used for DC charging.
- Communication: PLC for DC charging (ISO 15118, IEC 61851, DIN spec 70121) and PWM signaling for AC charging.
- Standard: Complies with IEC 62196.
- Usage: Widely accepted in Europe, India, and the UK, especially for high-voltage battery systems (up to 1000V, 500A, 400kW).
- **GB/T DC**:
- Origin: China.
- Pins: DC+, DC-, S+, S- (communication), A+, A- (auxiliary supply), CC1, CC2, PE.
- Communication: Uses CAN communication.
- Standard: Connector specifications in GB/T 20234.3.
- Voltage/Current: Supports up to 1500V and 800A.
- **BEVC-DC001 (Bharat Electric Vehicle DC Charging System)**:
- Origin: Department of Heavy Industries, Government of India.
- Power: Supports up to 10 kW for 48V systems and 15 kW for 72V systems (up to 200A).
- Communication: Uses CAN communication, derived from IEC 61851-24 and GB/T 27930 (application layer).
- Connector: Uses GB/T 20234.3 connector with a 5m cable.
- Usage: Primarily for low-voltage battery systems in early three-wheelers and e-cars in India.

- **DC Connector for LEVs (Type 6, India)**:
- Origin: Bureau of Indian Standards (BIS), incorporated in IS 17017-2-6.
- Power: Supports up to 12 kW for 120V DC (100A).
- Pins: DC+, DC-, AUX1, AUX2, COM1 (CAN high), COM2 (CAN low), CP.
- Communication: Uses CAN communication as per IS 17017-25.
- Usage: For light electric vehicles (LEVs) with low-voltage powertrains (e.g., two-wheelers, three-wheelers).
- **Combined AC and DC Connector for LEVs (Type 7, India)**:
- Origin: BIS, incorporated in IS 17017-2-7.
- Power: Supports up to 12 kW for 120V DC and 7 kW for 240V AC (32A).
- Pins: Neutral, Live, Earth, DC+, DC-, CP, PP, CAN+, CAN-.
- Communication: Uses CAN communication as per IS 17017-31.
- Usage: For LEVs, providing both AC and DC charging capability through a single connector.

6. Regional and Indian Charging Standards Overview

The global EV charging landscape shows regional preferences:
* **US**: Type 1 (AC), CCS1 (DC), CHAdeMO, and proprietary Tesla connectors.
* **Japan**: Type 1 (AC), CHAdeMO (DC).
* **Europe, UK, and most of the world**: Type 2 (AC), CCS2 (DC).
* **China**: GB/T AC, GB/T DC.

India presents a more complex and evolving scenario with a multitude of coexisting standards:
* **High-Voltage Powertrains (e.g., 4-wheelers)**: Primarily use CCS2 for DC charging and AC Type 2 for AC charging. These are incorporated into AIS 138 Part 1 (AC) and Part 2 (DC), and BIS IS 17017-2 Section 2 (AC Type 2) and Section 3 (DC, including CCS2 and CHAdeMO).
* **Low-Voltage Powertrains (e.g., 2-wheelers, 3-wheelers)**:
* **AC001**: An early standard for AC charging.
* **BEVC-DC001 (DC001)**: For DC charging of 48V/72V systems.
* **BIS IS 17017-2-6 (Type 6)**: A dedicated DC connector for LEVs.
* **BIS IS 17017-2-7 (Type 7)**: A combined AC and DC connector for LEVs.
* **Communication Standards**: India incorporates ISO 15118 for PLC communication (CCS2) and various CAN communication standards (IS 17017-24, -25, -31) for other systems like CHAdeMO and LEV connectors.

This diverse set of standards in India reflects the need to support a wide range of vehicle types, from low-voltage two-wheelers to high-voltage four-wheelers, and the ongoing transition in the EV ecosystem.

Key Takeaways

- [ ] EV charging is classified by modes (Mode 1-4) and power levels (Level 1-3), with Mode 4 and Level 3 DC representing fast charging.
- [ ] Public charging stations require robust communication between EVSEs, EVs (via BMS), and a Central Management System (CMS).
- [ ] AC charging relies on the vehicle's onboard charger, while DC charging involves power conversion within the EVSE for faster speeds.
- [ ] Different regions have adopted specific AC and DC connector standards, such as Type 1/CCS1 (US/Japan), Type 2/CCS2 (Europe/India), and GB/T (China).
- [ ] Combined Charging Systems (CCS1 and CCS2) allow both AC and DC charging through a single vehicle inlet, reducing complexity for manufacturers and users.
- [ ] India has a complex and evolving charging standard landscape, including AC001, DC001, AIS 138, and the BIS IS 17017 series, catering to both high and low-voltage EVs.
- [ ] Communication protocols like PLC (for CCS) and CAN (for CHAdeMO, GB/T, and some Indian standards) are crucial for safe and efficient DC charging.
- [ ] The EV charging ecosystem is dynamic, with continuous technological advancements and amendments to standards requiring ongoing updates.

Conclusion

The landscape of EV charging is characterized by a rich diversity of technologies and standards, each designed to meet specific regional needs and vehicle requirements. From the foundational modes and power levels that define charging capabilities to the intricate communication protocols that ensure safety and efficiency, every component plays a vital role. The emergence of combined charging systems like CCS1 and CCS2 highlights a global trend towards interoperability and user convenience, while countries like India are developing comprehensive frameworks to support a broad spectrum of electric vehicles, from two-wheelers to heavy-duty commercial vehicles. As the EV industry continues its rapid growth, the charging infrastructure will undoubtedly evolve further, with ongoing standardization efforts and technological innovations shaping a more seamless and accessible future for electric mobility. Staying informed about these developments is key to understanding the future of transportation.